

The Next 50 Years

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Abstract

The Next 50 Years is a special project designed for SIGGRAPH 2024. Following the 50th anniversary of the conference, it collected messages from a diverse group of participants, sharing their visions about the future of computer graphics and interactive techniques. This paper describes the process to gather and to showcase the contributions that were received. It employs media theory as a method to classify messages according to their communicative intention. Then, it discusses the development of a dedicated website that both exhibits the contributions and explores thematic interconnections and recurring topics such as artificial intelligence and generativity. In the end, the ongoing efforts on this project align with the contributors' visions for the future of technology, focusing on innovation, community engagement, and digital media practices.

CCS Concepts

• **Applied computing** → **Media arts**; • **Information systems** → **Web applications**; • **Human-centered computing** → **Visualization application domains**.

Keywords

Media and communication, Digital Messages, Community engagement, Future projections, Generativity

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1 Introduction

The Next 50 Years is a special project conceived for SIGGRAPH 2024 that follows a series of commemorations of the 50 years of the conference. In essence, it is a tribute to a renowned group of people who were initiators of the field of computer graphics and interactive techniques, but it is also a collective invitation to the community that has thrived throughout the years. We asked participants to share their vision in the form of a message about CGI in the forthcoming half of a century. In this paper we discuss the goals of the project, its theoretical foundations, and related work. Then, we introduce our methodology and technical strategies to gather and to represent the information we received. In the last

part, we comment on the value of the contributions for the field and the community.

2 Related work

Writing messages is one of the defining practices that we find in digital and interactive techniques. After the era of punched cards, the command line introduced a generalized manner to dialogue with a computer using text-based instructions. From the first programming languages and conversational chatbots there has been a deep fascination with the effect of seeing what we type on screen and its transformations with processing techniques. The field of electronic literature, for example, is particularly rich in creative and artistic examples, honoring its heritage from artistic movements such as Dada, Fluxus, and Futurism [Rettberg 2019].

In relation to participative experiences, the project “Messages vers le futur” was an experience we designed in 2012 which gave us valuable insights [Reyes 2022]. It consisted of a public installation exhibited at the event La Nuit de Chercheurs 2012 in France. Attendees to the event, logged into their Twitter account and wrote a message including a key hashtag. The message then appeared as 3D text on a large screen, traversing a hyperboloid shape to disappear slowly in the distance. The wormhole represented the idea of sending a message through time. The tweets were stored in an email account with the promise to be sent back to their authors within 10 years. This installation brought forward three aspects to consider about digital messages in long-term projects: the technological support of the message (how it is stored, in which format, how is it related to its original platform); the changes in the practices of messaging (new terms, new idioms, new signs); and, the comparative analysis of the content in the messages (what can we learn from them).

Our contribution can be distinguished from other projects that celebrate SIGGRAPH as a conference, as an actor of innovation, and as a hub where links are bounded between people. The SIGGRAPH 2018 booth, for example, was part of conference chair Roy Anthony’s vision for the voice of the generations (figure 1). For its part, [Kasik et al. 2023] collected memories of authors and committee members throughout the history of SIGGRAPH. Among the 200 stories they curated, we only found three participants in common with our project: Yoichiro Kawaguchi, Richard Chuang, and Alvy Ray Smith.

3 What is a message

The notion of “message” has been systematically studied in media and communication sciences, starting with the idea of quantifying the amount of information that was transmitted from one point to another through technological means. Mathematician Norbert Wiener described that messages were the subject of three simultaneous origins [Wiener 1948]. First, Wiener himself was interested

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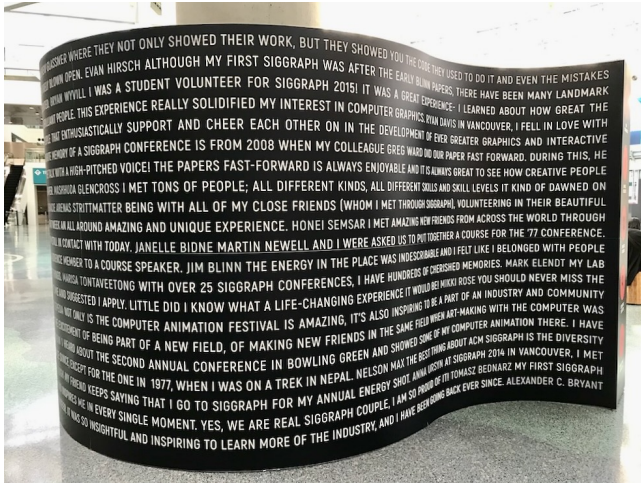


Figure 1: The SIGGRAPH 2018 booth at SIGGRAPH 2017 in Los Angeles. Photo courtesy of June Kim.

in communication mechanisms in technological and biological systems. Second, scientist and engineer Claude Shannon put an accent on coding information [Shannon 1948]. And third, statistician Ronald Fisher centered around the logarithmic distribution of data.

These ideas had an impact on humanities and social sciences. Linguist Roman Jakobson, who joined Harvard in 1949, elaborated his theory on the communicative functions in 1960 [Jakobson 2014]. He situated the “message” at the center of a process between a sender and a receiver, surrounded by the context, the channel, and the material aspect of the language. In this regard, a message could have different intentions according to the element which is emphasized. The process can be expressive (sender), imperative (receiver), phatic (channel), metalinguistic (language itself), referential (context), or poetic (message). It is often the case that a process reflects more than one dimension, whereas one of them has always a higher priority in a hierarchical scale.

In a closer relation to media studies, scholar Marshall McLuhan introduced in 1964 his celebrated motto “the medium is the message” [McLuhan 1964]. For him, the message of a medium was different from its content. A message was seen as an effect, a consequence or a new scale that is introduced by a medium. On the contrary, the content of a medium is always another medium, in the sense that the content of a film is the script or the book on which it is based. Furthermore, McLuhan considered media as extensions of the human being. As such, media can be found in the form of mass media, digital media, fashion, food, and urban design, to mention some types.

These approaches guided theoretical and practical considerations of our project. We were aware that asking people from diverse backgrounds to send messages about the next 50 years of CGI could be seen as broad and speculative. We anticipated that their visions would vary widely, taking on both technical, artistic, and societal forms. Thus, our framework gave us a method to classify messages according to their communicative intention (section 5).

Table 1: Contributors of The Next 50 Years

Name of contributor	Media format of the message
Roy Ascott, Paul Brown, Richard Chuang, Ernest Edmonds, Behnaz Farahi, Monika Fleischmann & Wolfgang Strauss, Richard Lowenberg, Roger Malina, Bill Seaman, Alvy Ray Smith	Text
Annick Bureau, Brenda Laurel, Charles Baldwin	Text and Image
Yoichiro Kawaguchi, Christa Sommerer & Laurent Mignon-neau, Valerie Wolf Gang, Joanna Zylinska	Video
Aaron Marcus, David Em	Zoom conversation
Nick Montfort	Generative poetry

4 Methodology

4.1 Gathering information

To select the participants of our project, we first looked at the survey that the conference marketing team distributes at the end of the conference. We focused on the question asking respondents to identify potential keynote speakers they would like to hear in future conferences. From the SIGGRAPH 2023 survey, we obtained 189 unique names from 307 answers. Not surprisingly, the list included people from academia and industry, from the private and the public sector. Overall, the most recurrent names were Jensen Huang (19 times), James Cameron (16 times), and Ed Catmull (13 times). All other suggested speakers received fewer than 10 mentions.

The following step was to send invitations to participate in our project. We used mainly public contact information available in institutional and personal web sites. In total, we sent one hundred invitations through email, LinkedIn, Twitter, Facebook, BlueSky, YouTube, and Substack. In a period of three months, we obtained 46 responses accepting or declining our invitation. If someone declined, we requested a recommendation for another potential participant.

We imagined The Next 50 Years as a project open to any language and media format. We elaborated an online form and a digital repository to receive contributions, respecting the ACM permission release form.

4.2 Representing information

By June 2024, we gathered 20 completed contributions, which corresponded to our initial expectations. Participants sent messages mainly in the form of text, image, and video. One participation was in the form of a generative text, written for the web in JavaScript, and two others were a conversation that was recorded on Zoom. All messages were received in English except for one, which was recorded in Japanese. Table 1 summarizes the contributors and media forms.

We created a dedicated website to showcase a preview of the contributions. We chose the GitHub platform to facilitate collaborative work and track the versioning of files. The initial template was inspired by CSS libraries but we customized our own styles based on Bootstrap 5 and JavaScript. The website is a single-page with a simple architecture. It is divided into three sections: an index of participants, the contributions, and a map of keywords. Each contribution has its own section and can be accessed and linked individually.

The contributions can be explored either through the index of names or by freely scrolling the page. In the bottom of the page, the map of keywords section is intended to serve as a navigation device to discover relationships between the messages. All messages were converted to text with the intention to perform simple semantic analysis. We assigned a list of five keywords to each contribution, and we organized them in six categories which were automatically suggested. We obtained a total of 64 different keywords. The map of keywords shown in figure 2 enables users to click on any text in a column to highlight related categories, people, and subtopics. The figure shows the result of clicking on the topic "Artificial Intelligence" in the list of keywords (right column).

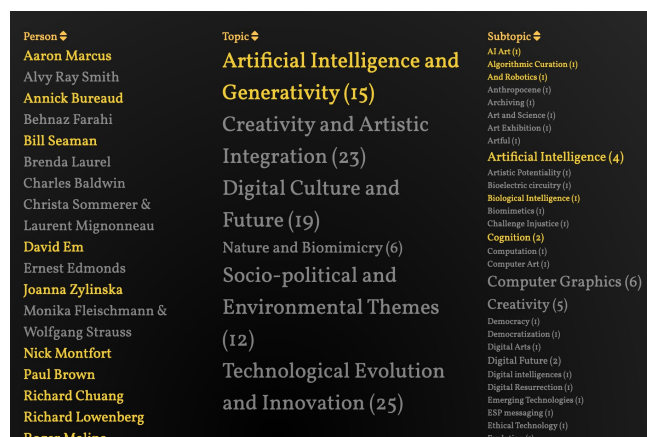


Figure 2: Map of keywords of the contributions.

The web application was displayed on site at SIGGRAPH 2024 as part of the Special Projects booth installed at the Colorado Convention Center (figure 3). The attendees of the conference were able to interact with the content and we observed different patterns. While some participants took notes, others captured photos of the screen. On average, a visitor stayed between 1 to 5 minutes, and many returned regularly. Others discovered more visualizations related to the current program of the conference and they used it as a guide and a conference planner.

5 Projecting the future

Our initial goal was to ask participants for their vision of the future of CGI and interactive techniques. Of course, talking about the future can be a difficult task. For some, it might be perceived as predicting, forecasting, or fortune-telling. However, in the words of media scholar Nick Montfort, the future is not something to be predicted, but something to be made [Montfort 2017]. The main



Figure 3: The Special Projects booth at SIGGRAPH 2024.

difference between predicting and projecting the future lies in the effort to plan how to achieve a vision. In this respect, the SIGGRAPH community has always attracted scientists, developers, inventors, writers, and educators who not only imagine new worlds and technologies, but also elaborate models, prototypes, sketches, blueprints, and workflows to reach a result. Among the respondents to our invitation, we count with early innovators in the field of computer graphics. This is inspiring because, seen at a distance, we are living in the world they imagined.

Recalling the communicative intention of the messages (section 3), the contributions assembled reflect on an expressive way to address the reader. Some are formulated as invitations, such as “join me” (Alvy Ray Smith) or “send me an email” (Bill Seaman), while others are calls to consider the critical times our world is going through (Behnaz Farahi, Monika Fleischmann and Richard Strauss). The contributions are also “referential” to the importance of the conference in the fields of computer science and the arts (Ernest Edmonds). Other messages put an accent on the language itself: they are “poetic” interpretations of the values that the conference represents, such as innovation, inclusiveness, and social commitment (Nick Montfort).

Among the topics present in the messages, it's unsurprising that "AI and generativity" were among the most frequently mentioned by our contributors. AI is seen as media because of the changes in scale it introduces. Aaron Marcus stressed that generating content has become so trivial that an increasing number of common users will produce massive amounts of new texts, images, and videos. For his part, David Em emphasized the importance of democratizing AI tools and discovering emerging artists from diverse cultures sharing their creations and experiences. Meanwhile, screen-based culture is continually enhanced by technologies that encourage us to reconsider how we self-curate and share generated content. As we adjust to these changes, the "content" of AI will likely include other media, such as simulations of fictional places, objects, and digital resurrection.

6 Further Work

The Next 50 Years is an experiment of collaborative interaction between the conference and its attendees. It explores new ways of capturing ideas about projecting possible yet attainable futures. By documenting, representing, and sharing messages from renowned participants, we aimed at emphasizing their importance to the community and the collective care for them.

The outcome of this first attempt is an interactive website, but it invites to imagine new forms. For example, June Kim, conference chair of SIGGRAPH Asia 2023, shared with us that she would envision an expansive concept involving multiple LED screens displaying a randomly selected message and photo from every member throughout the conference.

At the present moment, we expect that attendees and readers of this paper will feel inclined to participate in the project. We look forward to receiving messages from a diverse audience, enriching our list of keywords, and revealing emerging relationships between topics.

From a technical perspective, the project inscribes itself in a long tradition of dialogues between the artist, the public, the artwork, and the exhibition space. An enhancement to its current version could be a real-time semantic analysis of messages and an expanded interactivity with the public. However, we should also focus on how to effectively transmit messages to younger audiences and determine the appropriate format to use. Additionally, we should consider new strategies for web publishing, taking into account how the practices of texting and writing have shaped the expectations for digital tools. Lastly, we shall explore multidisciplinary perspectives for information processing, including how to preserve messages in the long-term.

From an aesthetic standpoint, our goal was to convey that imagining new futures is more productive when it is grounded on an individual and social well-being basis. As the history of technology

shows, our devices and artifacts can influence both positive and negative uses; they are not neutral, but neither do they determine our entire existence by themselves alone. Rather, they are part of a much larger network of cultural practices.

7 Conclusion

In this paper we discussed The Next 50 Years, a project specially conceived for SIGGRAPH 2024. We explored related works and theoretical backgrounds that were relevant to the production of the project. We described the methodology and results obtained so far. The project is an open work that aspires to attract new contributions from the community and to continue evolving its digital form. In its current version, it showcases 20 contributions in an interactive website. The contributors include individuals identified by the SIGGRAPH community as potential keynote speakers, but also contacts suggested by our respondents and participants who have expressed freely a willingness to be involved.

As the conference evolves, it is crucial to develop strategies to track its vast legacy and to think of desirable futures for its interaction with a society in constant transformation.

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